

ASSIGNMENT NO. 1: Introduction to Databases and SQL Environment

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Q1) Write SQL commands to create a database named SchoolDB.

The program should:

1. Create a new database named SchoolDB.
2. Select the database for use.
3. Display confirmation by creating a simple table inside it.

QUERY:

```
1 CREATE DATABASE schoolDB;
2 USE SchoolDB;
3 CREATE TABLE Student (
4     StudentID INT PRIMARY KEY,
5     StudentName VARCHAR(50),
6     Age INT,
7     ClassName VARCHAR(20),
8     city VARCHAR(30)
9 );
```

OUTPUT:

Your SQL query has been executed successfully.

[SHOW TABLES;](#)

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)] [[Refresh](#)]

Extra options

Tables_in_schooldb

student

Q2) Create a table named Student with the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Age	INT	
ClassName	VARCHAR(20)	
City	VARCHAR(30)	

The program should:

1. Create the table.
2. Define proper data types.
3. Apply primary key on StudentID.

QUERY:

```

1  USE SchoolDB;
2
3  DROP TABLE IF EXISTS Student;
4
5  CREATE TABLE Student (
6      StudentID INT PRIMARY KEY,
7      StudentName VARCHAR(50) NOT NULL,
8      Age INT,
9      ClassName VARCHAR(20),
10     City VARCHAR(30)
11 );
12
13 SHOW TABLES;|

```

OUTPUT:

Server: 127.0.0.1 / Database: schooldb

Structure SQL Search Query Export Import Operations Privileges Routines Events

```
DROP TABLE IF EXISTS Student;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0252 seconds.)

```
CREATE TABLE Student ( StudentID INT PRIMARY KEY, StudentName VARCHAR(50) NOT NULL, Age INT, ClassName VARCHAR(20), City VARCHAR(30) );
```

[Edit inline] [Edit] [Create PHP code]

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
SHOW TABLES;
```

☐ Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Tables_in_schooldb

student

Q3) Insert at least 10 records into the Student table.

The records should include:

1. Student ID
2. Student name
3. Age
4. Class name
5. City

After inserting data, display all records using the SELECT command.

QUERY:

```
1 INSERT INTO Student (StudentID, StudentName, Age, ClassName, City)
2 VALUES
3 (1, 'Aarav Shah', 18, 'FYBSc IT', 'Mumbai'),
4 (2, 'Meera Joshi', 19, 'FYBCA', 'Pune'),
5 (3, 'Rohan Patil', 20, 'SYBSc IT', 'Nashik'),
6 (4, 'Sneha Desai', 18, 'FYBCA', 'Mumbai'),
7 (5, 'Amit Kumar', 21, 'TYBSc IT', 'Delhi'),
8 (6, 'Pooja Singh', 19, 'SYBCA', 'Pune'),
9 (7, 'Vikram Rao', 20, 'SYBSc IT', 'Bengaluru'),
10 (8, 'Neha Sharma', 18, 'FYBCA', 'Nashik'),
11 (9, 'Rahul Verma', 21, 'TYBSc IT', 'Mumbai'),
12 (10, 'Kavita Iyer', 19, 'SYBCA', 'Chennai');
13
14 SELECT * FROM Student;
```

OUTPUT:

				StudentID	StudentName	Age	ClassName	City
☐	 Edit	 Copy	 Delete	1	Aarav Shah	18	FYBSc IT	Mumbai
☐	 Edit	 Copy	 Delete	2	Meera Joshi	19	FYBCA	Pune
☐	 Edit	 Copy	 Delete	3	Rohan Patil	20	SYBSc IT	Nashik
☐	 Edit	 Copy	 Delete	4	Sneha Desai	18	FYBCA	Mumbai
☐	 Edit	 Copy	 Delete	5	Amit Kumar	21	TYBSc IT	Delhi
☐	 Edit	 Copy	 Delete	6	Pooja Singh	19	SYBCA	Pune
☐	 Edit	 Copy	 Delete	7	Vikram Rao	20	SYBSc IT	Bengaluru
☐	 Edit	 Copy	 Delete	8	Neha Sharma	18	FYBCA	Nashik
☐	 Edit	 Copy	 Delete	9	Rahul Verma	21	TYBSc IT	Mumbai
☐	 Edit	 Copy	 Delete	10	Kavita Iyer	19	SYBCA	Chennai

Q4) Create a table named Course with the following fields:

Column Name	Data Type
CourseID	INT
CourseName	VARCHAR(50)
Duration	VARCHAR(20)
Fees	DECIMAL(10,2)

The program should:

1. Create the table.
2. Add a primary key on CourseID.
3. Insert at least 5 course records.
4. Display all course records.

QUERY:

```

1 USE SchoolDB;
2 CREATE TABLE Course (
3     CourseID INT PRIMARY KEY,
4     CourseName VARCHAR(50),
5     Duration VARCHAR(20),
6     Fees DECIMAL(10,2)
7 );
8 INSERT INTO Course (CourseID, CourseName, Duration, Fees)
9 VALUES
10 (101, 'Data Science', '3 Years', 75000.00),
11 (102, 'Computer Science', '3 Years', 70000.00),
12 (103, 'Cyber Security', '2 Years', 60000.00),
13 (104, 'Web Development', '1 Year', 45000.00),
14 (105, 'Artificial Intelligence', '2 Years', 80000.00);
15
16 SELECT * FROM Course;

```

OUTPUT:

		CourseID	CourseName	Duration	Fees
☐	Edit Copy Delete	101	Data Science	3 Years	75000.00
☐	Edit Copy Delete	102	Computer Science	3 Years	70000.00
☐	Edit Copy Delete	103	Cyber Security	2 Years	60000.00
☐	Edit Copy Delete	104	Web Development	1 Year	45000.00
☐	Edit Copy Delete	105	Artificial Intelligence	2 Years	80000.00

Q5) Write SQL queries to display only selected columns from the Student table.

Display:

1. StudentName
2. ClassName
3. City

Expected output should show only these three columns.

QUERY:

```

21
22 SELECT StudentName, ClassName, City
23 FROM Student;

```

OUTPUT:

				StudentName	ClassName	City
☐		Edit		Copy		Delete
				Aarav Shah	FYBSc IT	Mumbai
☐		Edit		Copy		Delete
				Meera Joshi	FYBCA	Pune
☐		Edit		Copy		Delete
				Rohan Patil	SYBSc IT	Nashik
☐		Edit		Copy		Delete
				Sneha Desai	FYBCA	Mumbai
☐		Edit		Copy		Delete
				Amit Kumar	TYBSc IT	Delhi
☐		Edit		Copy		Delete
				Pooja Singh	SYBCA	Pune
☐		Edit		Copy		Delete
				Vikram Rao	SYBSc IT	Bengaluru
☐		Edit		Copy		Delete
				Neha Sharma	FYBCA	Nashik
☐		Edit		Copy		Delete
				Rahul Verma	TYBSc IT	Mumbai
☐		Edit		Copy		Delete
				Kavita Iyer	SYBCA	Chennai

Q6) Write SQL commands to view the structure of the Student table. Use the suitable command according to the database environment:

Database	Command
MySQL	DESC Student;
PostgreSQL	student
SQLite	PRAGMA table_info(Student);

QUERY:

```
1 DESC Student;
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
StudentID	int(11)	NO	PRI	NULL	
StudentName	varchar(50)	NO		NULL	
Age	int(11)	YES		NULL	
ClassName	varchar(20)	YES		NULL	
City	varchar(30)	YES		NULL	

Q7) Prepare a CSV file named students.csv and import it into the Student table. The CSV file should contain:

StudentID	StudentName	Age	ClassName	City
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1	Aarav Shah	18	FYBSc IT	Mumbai
2	Meera Joshi	19	FYBCA	Pune
3	Rohan Patil	20	SYBSc IT	Nashik

The task should include:

1. Creating a matching table.
2. Importing the CSV file.
3. Displaying imported records using SELECT.

QUERY:

```

1 LOAD DATA INFILE "C:\\Users\\Acer\\Downloads\\students_new.csv"
2 INTO TABLE Student
3 FIELDS TERMINATED BY ','
4 ENCLOSED BY '"'
5 LINES TERMINATED BY '\n'
6 IGNORE 1 ROWS;
7
8 SELECT * FROM Student;

```

OUTPUT:

	StudentID	StudentName	Age	ClassName	City
☐ Edit Copy Delete	1	Aarav Shah	18	FYBSc IT	Mumbai
☐ Edit Copy Delete	2	Meera Joshi	19	FYBCA	Pune
☐ Edit Copy Delete	3	Rohan Patil	20	SYBSc IT	Nashik
☐ Edit Copy Delete	4	Sneha Desai	18	FYBCA	Mumbai
☐ Edit Copy Delete	5	Amit Kumar	21	TYBSc IT	Delhi
☐ Edit Copy Delete	6	Pooja Singh	19	SYBCA	Pune
☐ Edit Copy Delete	7	Vikram Rao	20	SYBSc IT	Bengaluru
☐ Edit Copy Delete	8	Neha Sharma	18	FYBCA	Nashik
☐ Edit Copy Delete	9	Rahul Verma	21	TYBSc IT	Mumbai
☐ Edit Copy Delete	10	Kavita Iyer	19	SYBCA	Chennai
☐ Edit Copy Delete	11	Ananya Kulkarni	19	FYBSc CS	Pune
☐ Edit Copy Delete	12	Aditya More	20	SYBCA	Mumbai
☐ Edit Copy Delete	13	Isha Gupta	18	FYBSc IT	Nagpur
☐ Edit Copy Delete	14	Kunal Patil	21	TYBSc CS	Nashik
☐ Edit Copy Delete	15	Riya Deshmukh	19	SYBSc IT	Kolhapur

Q8) Create a database named EmployeeDB and a table named Employee. The table should contain:

Column Name	Data Type
EmployeeID	INT

EmployeeName	VARCHAR(50)
Department	VARCHAR(30)
Salary	DECIMAL(10,2)
Column Name	Data Type
City	VARCHAR(30)

The program should:

1. Create the database.
2. Create the Employee table.
3. Insert 8 employee records.
4. Display all employee records.

QUERY:

```

1 CREATE DATABASE EmployeeDB;
2 USE EmployeeDB;
3 CREATE TABLE Employee (
4     EmployeeID INT PRIMARY KEY,
5     EmployeeName VARCHAR(50),
6     Department VARCHAR(30),
7     Salary DECIMAL(10,2),
8     City VARCHAR(30)
9 );
10 INSERT INTO Employee (EmployeeID, EmployeeName, Department, Salary, City)
11 VALUES
12 (1, 'Rahul Sharma', 'IT', 55000.00, 'Mumbai'),
13 (2, 'Priya Desai', 'HR', 48000.00, 'Pune'),
14 (3, 'Amit Patil', 'Finance', 62000.00, 'Nashik'),
15 (4, 'Sneha Joshi', 'IT', 58000.00, 'Nagpur'),
16 (5, 'Karan Mehta', 'Marketing', 45000.00, 'Mumbai'),
17 (6, 'Neha Kulkarni', 'HR', 50000.00, 'Pune'),
18 (7, 'Vikas Singh', 'Finance', 65000.00, 'Delhi'),
19 (8, 'Pooja Shah', 'IT', 60000.00, 'Bengaluru');
20
21 SELECT * FROM Employee;
```

OUTPUT:

		EmployeeID	EmployeeName	Department	Salary	City
☐	Edit Copy Delete	1	Rahul Sharma	IT	55000.00	Mumbai
☐	Edit Copy Delete	2	Priya Desai	HR	48000.00	Pune
☐	Edit Copy Delete	3	Amit Patil	Finance	62000.00	Nashik
☐	Edit Copy Delete	4	Sneha Joshi	IT	58000.00	Nagpur
☐	Edit Copy Delete	5	Karan Mehta	Marketing	45000.00	Mumbai
☐	Edit Copy Delete	6	Neha Kulkarni	HR	50000.00	Pune
☐	Edit Copy Delete	7	Vikas Singh	Finance	65000.00	Delhi
☐	Edit Copy Delete	8	Pooja Shah	IT	60000.00	Bengaluru

Q9) Write SQL queries to compare the difference between SELECT * and selecting specific columns.

The task should include:

1. Query using SELECT * FROM Student;
2. Query using SELECT StudentName, City FROM Student;

Write two lines explaining the difference between both outputs

QUERY:

```
1 SELECT * FROM Student;
2
3 SELECT StudentName, City FROM Student;
```

OUTPUT:

SELECT * FROM Student;

		StudentID	StudentName	Age	ClassName	City
☐	Edit Copy Delete	1	Aarav Shah	18	FYBSc IT	Mumbai
☐	Edit Copy Delete	2	Meera Joshi	19	FYBCA	Pune
☐	Edit Copy Delete	3	Rohan Patil	20	SYBSc IT	Nashik
☐	Edit Copy Delete	4	Sneha Desai	18	FYBCA	Mumbai
☐	Edit Copy Delete	5	Amit Kumar	21	TYBSc IT	Delhi
☐	Edit Copy Delete	6	Pooja Singh	19	SYBCA	Pune
☐	Edit Copy Delete	7	Vikram Rao	20	SYBSc IT	Bengaluru
☐	Edit Copy Delete	8	Neha Sharma	18	FYBCA	Nashik
☐	Edit Copy Delete	9	Rahul Verma	21	TYBSc IT	Mumbai
☐	Edit Copy Delete	10	Kavita Iyer	19	SYBCA	Chennai
☐	Edit Copy Delete	11	Ananya Kulkarni	19	FYBSc CS	Pune
☐	Edit Copy Delete	12	Aditya More	20	SYBCA	Mumbai
☐	Edit Copy Delete	13	Isha Gupta	18	FYBSc IT	Nagpur
☐	Edit Copy Delete	14	Kunal Patil	21	TYBSc CS	Nashik
☐	Edit Copy Delete	15	Riya Deshmukh	19	SYBSc IT	Kolhapur

SELECT StudentName, City FROM Student;

				StudentName	City
☐				Aarav Shah	Mumbai
☐				Meera Joshi	Pune
☐				Rohan Patil	Nashik
☐				Sneha Desai	Mumbai
☐				Amit Kumar	Delhi
☐				Pooja Singh	Pune
☐				Vikram Rao	Bengaluru
☐				Neha Sharma	Nashik
☐				Rahul Verma	Mumbai
☐				Kavita Iyer	Chennai
☐				Ananya Kulkarni	Pune
☐				Aditya More	Mumbai
☐				Isha Gupta	Nagpur
☐				Kunal Patil	Nashik
☐				Riya Deshmukh	Kolhapur

Two Line Explanation:

SELECT * displays **all columns and all records** from the Student table.

SELECT StudentName, City displays **only the specified columns**, so the output contains less information.

Q10) Create a complete SQL script for a mini college database.

The script should include:

1. Creating database MiniCollegeDB.
2. Creating tables Student, Course, and Faculty.
3. Adding suitable columns and data types.
4. Inserting at least 5 records in each table.
5. Displaying all records from each table.
6. Viewing the structure of each table.

QUERY:

```
1 CREATE DATABASE MiniCollegeDB;
2 USE MiniCollegeDB;
3
4 CREATE TABLE Student(
5 StudentID INT PRIMARY KEY, StudentName VARCHAR(50), Age INT);
6
7 CREATE TABLE Course(
8 CourseID INT PRIMARY KEY, CourseName VARCHAR(50), Fees DECIMAL(10,2));
9
10 CREATE TABLE Faculty(
11 FacultyID INT PRIMARY KEY, FacultyName VARCHAR(50), Department VARCHAR(30));
12
13 INSERT INTO Student VALUES
14 (1, 'Aarav', 18), (2, 'Meera', 19), (3, 'Rohan', 20), (4, 'Sneha', 18), (5, 'Amit', 21);
15
16 INSERT INTO Course VALUES
17 (101, 'Data Science', 75000), (102, 'Computer Science', 70000),
18 (103, 'Cyber Security', 60000), (104, 'Web Development', 45000), (105, 'AI', 80000);
19
20 INSERT INTO Faculty VALUES
21 (201, 'Rahul', 'IT'), (202, 'Priya', 'HR'), (203, 'Amit', 'Finance'),
22 (204, 'Neha', 'IT'), (205, 'Karan', 'Science');
23
24 SELECT * FROM Student;
25 SELECT * FROM Course;
26 SELECT * FROM Faculty;
27
28 DESC Student;
29 DESC Course;
30 DESC Faculty;
```

OUTPUT:

Output 1 — Student Table

		StudentID	StudentName	Age
☐	 Edit  Copy  Delete	1	Aarav	18
☐	 Edit  Copy  Delete	2	Meera	19
☐	 Edit  Copy  Delete	3	Rohan	20
☐	 Edit  Copy  Delete	4	Sneha	18
☐	 Edit  Copy  Delete	5	Amit	21

Output 2 — Course Table

		CourseID	CourseName	Fees
☐	 Edit  Copy  Delete	101	Data Science	75000.00
☐	 Edit  Copy  Delete	102	Computer Science	70000.00
☐	 Edit  Copy  Delete	103	Cyber Security	60000.00
☐	 Edit  Copy  Delete	104	Web Development	45000.00
☐	 Edit  Copy  Delete	105	AI	80000.00

Output 3 — Faculty Table

		FacultyID	FacultyName	Department
☐	 Edit  Copy  Delete	201	Rahul	IT
☐	 Edit  Copy  Delete	202	Priya	HR
☐	 Edit  Copy  Delete	203	Amit	Finance
☐	 Edit  Copy  Delete	204	Neha	IT
☐	 Edit  Copy  Delete	205	Karan	Science


☐ Check all
 With selected:
  Edit
  Copy
  Delete
  Export

Structure — Student

DESC Student;

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

Extra options

Field	Type	Null	Key	Default	Extra
StudentID	int(11)	NO	PRI	NULL	
StudentName	varchar(50)	YES		NULL	
Age	int(11)	YES		NULL	

Structure — Course

DESC Course;

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

Extra options

Field	Type	Null	Key	Default	Extra
CourseID	int(11)	NO	PRI	NULL	
CourseName	varchar(50)	YES		NULL	
Fees	decimal(10,2)	YES		NULL	

Structure — Faculty

DESC Faculty;

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

Extra options

Field	Type	Null	Key	Default	Extra
FacultyID	int(11)	NO	PRI	NULL	
FacultyName	varchar(50)	YES		NULL	
Department	varchar(30)	YES		NULL	

Result:

MiniCollegeDB was created successfully with Student, Course, and Faculty tables, and the required records and table structures were displayed.